

Archive: Removed Results

(mean queue lengths, mean waiting times, and loss fractions;
removed from the main thesis and preserved verbatim)

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This file collects material removed from the main thesis to keep the focus on the generating-function structure: the per-model mean-queue-length and mean-waiting-time derivations (Sections 1–5), and the loss-fraction / cost-of-abandonment analysis (Section 6). Everything is reproduced unchanged. Cross-references to the main document do not resolve in this standalone file.

1 Model-A (baseline, $\gamma_1 = \gamma_2 = 0, \theta_1 = \theta_2 = 0$)

1.1 Mean queue lengths and mean waiting times

The results for Model-A are well-studied and standard. However, they will here appear in a slightly different way compared to what usually appears on textbooks —e.g. Adan & Resing [?], since our states are defined with the number of customers in the *queues* with a busy server, as opposed to those in the system. The customer in service thus contributes to all those with a factor of ρ to the standard results we would typically find.

1.1.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (??) gives

$$[-\lambda_1(x-1)(x-1/\rho_1)]P(x,1) = \mu(x-1)P_y(1).$$

Cancelling the common factor $(x-1)$ for $x \neq 1$ and rearranging:

$$P(x,1) = \frac{\mu P_y(1)}{\mu - \lambda_1 x}.$$

The constant $P_y(1)$ is obtained using normalisation for all busy states $P(1,1) = 1 - \pi_0 = \rho$:

$$\rho = \frac{\mu P_y(1)}{\mu - \lambda_1} \implies P_y(1) = \rho(1 - \rho_1),$$

so that

$$P(x,1) = \frac{\rho(1 - \rho_1)}{1 - \rho_1 x}.$$

Differentiating by the quotient rule and evaluating at $x = 1$:

$$\mathbb{E}[N_1] = \left. \frac{d}{dx} P(x,1) \right|_{x=1} = \frac{\rho(1 - \rho_1) \cdot \rho_1}{(1 - \rho_1 x)^2} \Big|_{x=1} = \frac{\rho(1 - \rho_1) \rho_1}{(1 - \rho_1)^2} = \frac{\rho \rho_1}{1 - \rho_1}. \quad (1.1)$$

1.1.2 Total queue length from the diagonal

Setting $x = y = z$ in the kernel polynomial $\lambda_1 x^2 - (\lambda_1 + \lambda_2 + \mu)x + \mu$ shows that $x^*(z) = z$ iff $z \in \{1, 1/\rho\}$; hence $x^*(z) \neq z$ for all $z \in (0, 1)$ with $\rho < 1$. Therefore the ratio $(z - x^*(z))/(x^*(z) - z) = -1$ throughout $(0, 1)$, and the formula of Theorem ?? reduces to:

$$P(z, z) = \frac{z(1 - z) \rho(1 - \rho)}{D(z, z)} \cdot (-1).$$

The denominator at $x = y = z$ is $D(z, z) = (1 + \rho)z^2 - z - \rho z^3 = -\rho z(z - 1)(z - 1/\rho)$, so:

$$\begin{aligned} P(z, z) &= \frac{-z(1 - z) \rho(1 - \rho)}{-\rho z(z - 1)(z - 1/\rho)} \\ &= \frac{(1 - z)(1 - \rho)}{(z - 1)(z - 1/\rho)} = \frac{-(1 - \rho)}{z - 1/\rho} = \frac{\rho(1 - \rho)}{1 - \rho z}, \end{aligned} \quad (1.2)$$

confirming the diagonal identity of Corollary ??. Since $\frac{d}{dz} P(z, z) \Big|_{z=1} = \frac{\partial}{\partial x} P(1, 1) + \frac{\partial}{\partial y} P(1, 1) = \mathbb{E}[N_1] + \mathbb{E}[N_2]$, differentiating by the quotient rule and evaluating at $z = 1$ gives:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] = \left. \frac{d}{dz} \frac{\rho(1 - \rho)}{1 - \rho z} \right|_{z=1} = \frac{\rho(1 - \rho) \cdot \rho}{(1 - \rho z)^2} \Big|_{z=1} = \frac{\rho^2(1 - \rho)}{(1 - \rho)^2} = \frac{\rho^2}{1 - \rho}.$$

This is the $M/M/1(\lambda, \mu)$ queue-length formula for the combined system [?], as expected: the two classes together behave as a single $M/M/1$ queue.

1.1.3 Class-2 mean queue length

By subtracting the marginal for class-1 to the total number of customers, we find the marginal for class-2:

$$\begin{aligned} \mathbb{E}[N_2] &= (\mathbb{E}[N_1] + \mathbb{E}[N_2]) - \mathbb{E}[N_1] = \frac{\rho^2}{1 - \rho} - \frac{\rho \rho_1}{1 - \rho_1} \\ &= \frac{\rho^2(1 - \rho_1) - \rho \rho_1(1 - \rho)}{(1 - \rho)(1 - \rho_1)} = \frac{\rho[\rho(1 - \rho_1) - \rho_1(1 - \rho)]}{(1 - \rho)(1 - \rho_1)} \\ &= \frac{\rho(\rho - \rho_1)}{(1 - \rho)(1 - \rho_1)} = \frac{\rho \rho_2}{(1 - \rho_1)(1 - \rho)}. \end{aligned} \quad (1.3)$$

1.1.4 Mean waiting times via Little's Law

To obtain the expected waiting times, we can simply apply *Little's Law* ?? to each class independently: we then have $\mathbb{E}[N_i] = \lambda_i \mathbb{E}[W_i]$ where $\mathbb{E}[W_i]$ is the mean waiting time in the queue, excluding the time in service. Therefore:

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{\rho \rho_1}{\lambda_1(1 - \rho_1)} = \frac{\rho \cdot (\lambda_1/\mu)}{\lambda_1(1 - \rho_1)} = \frac{\rho}{\mu(1 - \rho_1)}, \quad (1.4)$$

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2} = \frac{\rho \rho_2}{\lambda_2(1 - \rho_1)(1 - \rho)} = \frac{\rho \cdot (\lambda_2/\mu)}{\lambda_2(1 - \rho_1)(1 - \rho)} = \frac{\rho}{\mu(1 - \rho_1)(1 - \rho)}. \quad (1.5)$$

As expected, class-1 benefits from the priority discipline, and $\mathbb{E}[W_2]$ carries the additional penalty factor $1/(1 - \rho)$ compared to $\mathbb{E}[W_1]$.

2 Model- B_2 (one-way jockeying, $\gamma_1 > 0$, $\gamma_2 = 0$)

2.1 Mean queue lengths and mean waiting times

Total queue length from the diagonal

Setting $x = y = z$ in the reduced fundamental equation (??), the jockeying convection term $\gamma_1 xy(x-y)\partial_x P$ vanishes at $x = y = z$ (factor $(x-y) = 0$), as does the boundary term $\mu(x-y)P_y(y)$. The remaining balance is identical to Model-A's, so the same diagonal cancellation gives

$$P(z, z) = \frac{\rho(1 - \rho)}{1 - \rho z},$$

the same formula as (1.2). Since $P(z, z) = \mathbb{E}[z^{N_1+N_2}] = \mathbb{E}[z^N]$ is the PGF of the total in-queue count N , the generating-function identity $\frac{d}{dz} P(z, z)|_{z=1} = \mathbb{E}[N_1] + \mathbb{E}[N_2]$ holds; differentiating by the quotient rule and evaluating at $z = 1$:

$$\begin{aligned} \mathbb{E}[N_1] + \mathbb{E}[N_2] &= \left. \frac{d}{dz} \frac{\rho(1 - \rho)}{1 - \rho z} \right|_{z=1} = \left. \frac{\rho(1 - \rho) \rho}{(1 - \rho z)^2} \right|_{z=1} \\ &= \frac{\rho^2(1 - \rho)}{(1 - \rho)^2} = \frac{\rho^2}{1 - \rho}. \end{aligned}$$

One-way jockeying does not alter the total mean queue length relative to Model-A, since the diagonal PGF is unchanged.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}|_{(1,1)}$ by first reducing the fundamental equation (??) at $y = 1$, and then differentiating the resulting marginal PGF at $x = 1$.

Step 1: marginal PGF $P(x, 1)$. Evaluating (??) at $y = 1$, the bracket multiplying P becomes $(\lambda_1 + \mu)x - \mu - \lambda_1 x^2 = -\lambda_1(x - 1)(x - 1/\rho_1)$, the convection coefficient is $\gamma_1 x(x - 1)$, and the right-hand side reduces to $\mu(x - 1)P_y(1)$ (the $\pi(0, 0)$ term carries the factor $(1 - y) = 0$). Every term therefore shares the common factor $(x - 1)$; cancelling it leaves the first-order linear ODE in x :

$$-\lambda_1(x - 1/\rho_1) P(x, 1) + \gamma_1 x \frac{d}{dx} P(x, 1) = \mu P_y(1).$$

The integrating factor for this ODE is $e^{-\lambda_1 x/\gamma_1} x^{\mu/\gamma_1}$ (noting $\lambda_1/\rho_1 = \mu$), and integrating from 0 to x (the boundary term vanishes since $0^{\mu/\gamma_1} = 0$):

$$P(x, 1) e^{-\lambda_1 x/\gamma_1} x^{\mu/\gamma_1} = \frac{\mu P_y(1)}{\gamma_1} \int_0^x e^{-\lambda_1 t/\gamma_1} t^{\mu/\gamma_1 - 1} dt.$$

Rearranging and writing $\alpha = \mu/\gamma_1$ and $\mathcal{H}(x) = \int_0^x e^{-\lambda_1 t/\gamma_1} t^{\alpha - 1} dt$:

$$P(x, 1) = \frac{\mu P_y(1)}{\gamma_1} e^{\lambda_1 x/\gamma_1} x^{-\alpha} \mathcal{H}(x).$$

The constant $P_y(1)$ follows from $P(1, 1) = 1 - \pi_0 = \rho$, which gives $P_y(1) = \rho \gamma_1 / (\mu e^{\lambda_1/\gamma_1} \mathcal{H}(1))$.

Step 2: differentiate $P(x, 1)$ at $x = 1$. Writing $P(x, 1) = C \cdot f(x) \cdot \mathcal{H}(x)$ with $C = \mu P_y(1)/\gamma_1$ and $f(x) = e^{\lambda_1 x/\gamma_1} x^{-\alpha}$, the product rule together with Leibniz's rule ($d\mathcal{H}/dx = e^{-\lambda_1 x/\gamma_1} x^{\alpha-1}$, the integrand at the upper limit) gives

$$\frac{d}{dx} P(x, 1) = C \left[f'(x) \mathcal{H}(x) + f(x) e^{-\lambda_1 x/\gamma_1} x^{\alpha-1} \right].$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\gamma_1 - \alpha/x = (\lambda_1 x - \mu)/(\gamma_1 x)$, and the second product term simplifies as $f(x) e^{-\lambda_1 x/\gamma_1} x^{\alpha-1} = x^{-1}$, so

$$\frac{d}{dx} P(x, 1) = \frac{\mu P_y(1)}{\gamma_1} \left[e^{\lambda_1 x/\gamma_1} x^{-\alpha} \frac{\lambda_1 x - \mu}{\gamma_1 x} \mathcal{H}(x) + x^{-1} \right].$$

Step 3: evaluate at $x = 1$.

$$\begin{aligned} \mathbb{E}[N_1] &= \left. \frac{d}{dx} P(x, 1) \right|_{x=1} \\ &= \frac{\mu P_y(1)}{\gamma_1} \left[e^{\lambda_1/\gamma_1} \frac{\lambda_1 - \mu}{\gamma_1} \mathcal{H}(1) + 1 \right], \end{aligned} \quad (2.1)$$

where $\mathcal{H}(1) = \int_0^1 e^{-\lambda_1 t/\gamma_1} t^{\mu/\gamma_1-1} dt = (\gamma_1/\lambda_1)^{\mu/\gamma_1} \gamma(\mu/\gamma_1, \lambda_1/\gamma_1)$ and $\gamma(\cdot, \cdot)$ is the lower incomplete gamma function. The factor $(\lambda_1 - \mu) < 0$ (since $\rho_1 = \lambda_1/\mu < 1$), confirming that $e^{\lambda_1/\gamma_1}(\lambda_1 - \mu)\mathcal{H}(1)/\gamma_1 + 1 > 0$. No elementary simplification of (2.1) is known; the expression is evaluated numerically. As $\gamma_1 \rightarrow 0^+$ the Model-A value (1.1) is recovered.

Class-2 mean queue length by subtraction

$$\mathbb{E}[N_2] = (\mathbb{E}[N_1] + \mathbb{E}[N_2]) - \mathbb{E}[N_1] = \frac{\rho^2}{1 - \rho} - \mathbb{E}[N_1].$$

Since jockeying defections reduce the class-1 queue, $\mathbb{E}[N_1]$ is smaller than the Model-A value $\rho\rho_1/(1 - \rho_1)$, so $\mathbb{E}[N_2]$ correspondingly exceeds the Model-A value.

Mean waiting times via Little's Law

Little's Law applies here without qualification. It is a rate-conservation identity — for any subsystem in which customers are neither created nor destroyed internally, the time-average number present equals the long-run entry rate times the mean time an entrant spends inside — and it holds for *any* positive-recurrent system with finite mean queue length (here ensured by $\rho < 1$, which gives $\mathbb{E}[N_1] \leq \mathbb{E}[N] = \rho^2/(1 - \rho) < \infty$ and so makes the jockeying rate $\gamma_1 \mathbb{E}[N_1]$ below a genuine finite rate), requiring neither Poisson input nor independence. In particular it remains valid even though one-way jockeying has destroyed the per-class Poisson/renewal structure (the very obstruction that ruled out a probabilistic route in Step (ii)). We apply it to each queue (the waiting line, excluding the customer in service), with $\mathbb{E}[W_i]$ understood as the mean time a customer spends in Queue i averaged over all customers that enter it — an arrival routed directly into service contributing a zero-length visit.

Class 1. Every class-1 arrival is counted as an entrant to Queue 1 at rate λ_1 (those finding the server idle enter service at once, contributing zero waiting time), and each leaves Queue 1 either by entering service or by jockeying to Queue 2. Since no other stream feeds Queue 1, its throughput is exactly λ_1 , and Little's Law gives

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1}. \quad (2.2)$$

Class 2. Queue 2 has *two* input streams: the class-2 Poisson arrivals at rate λ_2 and the jockeying transfers from Queue 1. Because each of the N_1 waiting class-1 customers jockeys independently at rate γ_1 , the long-run transfer rate is $\gamma_1 \mathbb{E}[N_1]$ (a genuine rate by the ergodic theorem, equal to the time average of the instantaneous jockeying intensity $\gamma_1 N_1$). The throughput of Queue 2 is therefore $\lambda_2 + \gamma_1 \mathbb{E}[N_1]$, and Little's Law reads $\mathbb{E}[N_2] = (\lambda_2 + \gamma_1 \mathbb{E}[N_1]) \mathbb{E}[W_2]$, so that

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2 + \gamma_1 \mathbb{E}[N_1]}. \quad (2.3)$$

Here $\mathbb{E}[W_2]$ is the mean time in the class-2 line averaged over *all* Queue-2 entrants of both streams; a jockeyed customer's Queue-2 clock starts at the transfer instant, so its total wait is an initial

Queue-1 segment (of mean $\mathbb{E}[W_1]$) followed by this Queue-2 segment. Thus (2.3) is the per-Queue-2-entrant residence time, not the end-to-end delay of a tagged class-2 *arrival*.

The jockeying inflow is what distinguishes (2.3) from the single-stream Models A and C_2 : the naive denominator λ_2 would undercount the entrants to Queue 2 and is correct only in the no-jockeying limit. Indeed, as $\gamma_1 \rightarrow 0^+$ the transfer rate $\gamma_1 \mathbb{E}[N_1] \rightarrow 0$, so (2.3) reduces to $\mathbb{E}[N_2]/\lambda_2$ and both (2.2) and (2.3) recover the Model- A values (1.4)–(1.5).

3 Model- C_2 (class-1 abandonment, $\theta_1 > 0$)

3.1 Mean queue lengths and mean waiting times

Why the diagonal no longer simplifies

Setting $x = y = z$ in (??), the derivative term of abandonment $\theta_1 xy(x-1) \partial_x P$ survives (the factor $(x-1)|_{x=z} \neq 0$ for $z \in (0, 1)$), and the diagonal equation reads

$$[(\lambda_1 + \lambda_2 + \mu)z^2 - \mu z - \lambda z^3]P(z, z) + \theta_1 z^2(z-1) \frac{\partial P}{\partial x}(z, z) = -\mu z(1-z) \pi(0, 0). \quad (3.1)$$

The contrast with Model- A —or any model without abandonment—is instructive: $P(z, z)$ is no longer purely algebraic with closed form $P(z, z) = \pi(0, 0)/(1 - \rho z)$ (1.2), from which the total $\mathbb{E}[N_1] + \mathbb{E}[N_2]$ followed by a single differentiation. Here the surviving derivative term remains as a second unknown, so that route is no longer available; abandonment removes customers and lowers both means relative to models without it.

Class-2 mean queue length via $P(1, y)$

Setting $x = 1$ in (??) we obtain

$$\lambda_2 y(1-y) P(1, y) = \mu(1-y)[P_y(y) - \pi(0, 0)].$$

Cancelling $(1-y)$ for $y \neq 1$ and substituting the known $P_y(y)$ (??),

$$P(1, y) = \frac{\mu[P_y(y) - \pi(0, 0)]}{\lambda_2 y} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{1 - \tilde{B}_C(\lambda_2(1-y))}{\tilde{B}_C(\lambda_2(1-y)) - y},$$

the marginal PGF of class-2. It is 0/0 at $y = 1$, so set $\varepsilon = 1 - y \rightarrow 0$ and use the LST Taylor expansion $\tilde{B}_C(\lambda_2 \varepsilon) = 1 - \lambda_2 \mathbb{E}[B_C] \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3)$:

$$\begin{aligned} 1 - \tilde{B}_C(\lambda_2 \varepsilon) &= \lambda_2 \mathbb{E}[B_C] \varepsilon - \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3), \\ \tilde{B}_C(\lambda_2 \varepsilon) - y &= (1 - \lambda_2 \mathbb{E}[B_C] \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2) - (1 - \varepsilon) = (1 - \lambda_2 \mathbb{E}[B_C]) \varepsilon + \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2] \varepsilon^2 + O(\varepsilon^3). \end{aligned}$$

Writing $a = \lambda_2 \mathbb{E}[B_C]$ and $c = \frac{1}{2} \lambda_2^2 \mathbb{E}[B_C^2]$,

$$P(1, 1 - \varepsilon) = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{a\varepsilon - c\varepsilon^2 + O(\varepsilon^3)}{(1-a)\varepsilon + c\varepsilon^2 + O(\varepsilon^3)} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{a - c\varepsilon}{(1-a) + c\varepsilon} + O(\varepsilon^2).$$

Differentiating $f(\varepsilon) = (a - c\varepsilon)/((1-a) + c\varepsilon)$ by the quotient rule,

$$f'(\varepsilon) = \frac{-c((1-a) + c\varepsilon) - (a - c\varepsilon)c}{((1-a) + c\varepsilon)^2} = \frac{-c}{(1-a + c\varepsilon)^2},$$

and since $\frac{d}{dy} P(1, y)|_{y=1} = -\frac{d}{d\varepsilon} P(1, 1 - \varepsilon)|_{\varepsilon=0}$,

$$\mathbb{E}[N_2] = \frac{d}{dy} P(1, y) \Big|_{y=1} = \frac{\mu \pi(0, 0)}{\lambda_2} \cdot \frac{c}{(1-a)^2} = \frac{\mu \pi(0, 0) \lambda_2 \mathbb{E}[B_C^2]}{2(1 - \lambda_2 \mathbb{E}[B_C])^2}. \quad (3.2)$$

As $\theta_1 \rightarrow 0^+$, $\mathbb{E}[B_C] \rightarrow (\mu - \lambda_1)^{-1}$ and $\mathbb{E}[B_C^2] \rightarrow 2\mu/(\mu - \lambda_1)^3$, recovering the Model- A value $\mathbb{E}[N_2] = \rho \rho_2 / ((1 - \rho_1)(1 - \rho))$.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x} \Big|_{(1,1)}$ by evaluating the theorem formula at $y = 1$ and then differentiating in x .

Step 1: marginal PGF $P(x, 1)$. As $y \rightarrow 1$, the exponent $\beta(y) \rightarrow 0$, so $(1-x)^{\beta(y)} \rightarrow 1$ for fixed $x \in (0, 1)$, and $\mathcal{H}_1(x, y) \rightarrow \int_0^x e^{-\lambda_1 t/\theta_1} t^{\mu/\theta_1-1} dt$ while $\mathcal{H}_2(x, y)$ tends to a finite limit. In the theorem's bracket the two integrals carry different weights: $\tilde{B}_C(\lambda_2(1-y)) \rightarrow 1$, while $1 - \tilde{B}_C(\lambda_2(1-y)) = \lambda_2 \mathbb{E}[B_C](1-y) + O((1-y)^2) \rightarrow 0$, so the $\mathcal{H}_2$ term drops and the bracket collapses to $\mathcal{H}_1(x, 1)$. The prefactor's y -dependent part satisfies $(1-y)/(\tilde{B}_C(\lambda_2(1-y)) - y) \rightarrow 1/(1 - \lambda_2 \mathbb{E}[B_C])$, so

$$P(x, 1) = \frac{\mu \pi(0, 0)}{\theta_1 (1 - \lambda_2 \mathbb{E}[B_C])} e^{\lambda_1 x/\theta_1} x^{-\mu/\theta_1} \int_0^x e^{-\lambda_1 t/\theta_1} t^{\mu/\theta_1-1} dt.$$

Step 2: differentiate $P(x, 1)$ at $x = 1$. Write $P(x, 1) = C f(x) \mathcal{K}(x)$ with $C = \mu \pi(0, 0)/(\theta_1 (1 - \lambda_2 \mathbb{E}[B_C]))$, $f(x) = e^{\lambda_1 x/\theta_1} x^{-\mu/\theta_1}$ and $\mathcal{K}(x) = \int_0^x e^{-\lambda_1 t/\theta_1} t^{\mu/\theta_1-1} dt$. By the product rule with Leibniz's rule ($\mathcal{K}'(x) = e^{-\lambda_1 x/\theta_1} x^{\mu/\theta_1-1}$) and the logarithmic derivative $f'(x)/f(x) = (\lambda_1 x - \mu)/(\theta_1 x)$,

$$\frac{d}{dx} P(x, 1) = C \left[f'(x) \mathcal{K}(x) + f(x) e^{-\lambda_1 x/\theta_1} x^{\mu/\theta_1-1} \right] = C \left[e^{\lambda_1 x/\theta_1} x^{-\mu/\theta_1} \frac{\lambda_1 x - \mu}{\theta_1 x} \mathcal{K}(x) + x^{-1} \right].$$

Step 3: evaluate at $x = 1$.

$$\mathbb{E}[N_1] = \frac{d}{dx} P(x, 1) \Big|_{x=1} = \frac{\mu \pi(0, 0)}{\theta_1 (1 - \lambda_2 \mathbb{E}[B_C])} \left[e^{\lambda_1/\theta_1} \frac{\lambda_1 - \mu}{\theta_1} \mathcal{K}(1) + 1 \right],$$

where $\mathcal{K}(1) = \int_0^1 e^{-\lambda_1 t/\theta_1} t^{\mu/\theta_1-1} dt = (\theta_1/\lambda_1)^{\mu/\theta_1} \gamma(\mu/\theta_1, \lambda_1/\theta_1)$, with $\gamma(\cdot, \cdot)$ the lower incomplete gamma function. Since $\lambda_1 - \mu < 0$ ($\rho_1 < 1$) the bracket is positive. No elementary simplification is known, so $\mathbb{E}[N_1]$ is evaluated numerically; as $\theta_1 \rightarrow 0^+$ it recovers the Model-A value (1.1).

Mean waiting times via Little's Law

The relation $\mathbb{E}[N_i] = \lambda_i \mathbb{E}[W_i]$ holds for any stable system. Here λ_i is the arrival rate of *all* class- i customers, including those that eventually abandon, so $\mathbb{E}[W_i]$ is the mean queueing time averaged over both served and abandoned customers:

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1}, \quad \mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2}.$$

Using the throughput $\mu(1-\pi_0)$ instead would give the conditional waiting time of served customers only—a different object. As $\theta_1 \rightarrow 0^+$ both means recover the Model-A values (1.1)–(1.5).

4 Model- B_2^H (head-of-line jockeying)

4.1 Mean queue lengths and mean waiting times

4.1.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (??), the term $\mu x(1-y)\pi(0, 0)$ vanishes and the coefficient of $P(x, 1)$ factors as $-(x-1)(\lambda_1 x - (\mu + \gamma_1))$, while the right-hand side becomes $(x-1)(\mu + \gamma_1)P_y(1)$. Cancelling $(x-1)$ for $x \neq 1$,

$$P(x, 1) = \frac{(\mu + \gamma_1) P_y(1)}{\mu + \gamma_1 - \lambda_1 x} = \frac{P_y(1)}{1 - \rho_B x}, \quad \rho_B := \frac{\lambda_1}{\mu + \gamma_1}. \quad (4.1)$$

This is a geometric class-1 marginal: the priority queue behaves as an $M/M/1$ queue whose effective *clearance rate* is $\mu + \gamma_1$, since a waiting class-1 customer at the head of Queue 1 departs that queue either by entering service (rate μ) or by jockeying (rate γ_1). The constant $P_y(1)$ is pinned by the normalisation $P(1, 1) = 1 - \pi_0 = \rho$, giving $P_y(1) = \rho(1 - \rho_B)$, so that $P(x, 1) = \rho(1 - \rho_B)/(1 - \rho_B x)$. Differentiating by the quotient rule and evaluating at $x = 1$,

$$\mathbb{E}[N_1] = \frac{d}{dx} P(x, 1) \Big|_{x=1} = \frac{\rho(1 - \rho_B) \rho_B}{(1 - \rho_B)^2} = \frac{\rho \rho_B}{1 - \rho_B}. \quad (4.2)$$

This is Model-A's formula (1.1) with the priority load ρ_1 replaced by the *jockeying-reduced* load $\rho_B = \lambda_1/(\mu + \gamma_1) \leq \rho_1$.

4.1.2 Total queue length from the diagonal

Because jockeying conserves the total number of customers, the combined process $N_1 + N_2$ (together with the in-service customer) is the queue-length process of an ordinary $M/M/1(\lambda, \mu)$ queue, and the diagonal generating function (??) is $P(z, z) = \rho(1 - \rho)/(1 - \rho z)$. Since $\frac{d}{dz}P(z, z)|_{z=1} = \mathbb{E}[N_1] + \mathbb{E}[N_2]$, differentiating gives

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] = \frac{d}{dz} \frac{\rho(1 - \rho)}{1 - \rho z} \Big|_{z=1} = \frac{\rho^2(1 - \rho)}{(1 - \rho)^2} = \frac{\rho^2}{1 - \rho}, \quad (4.3)$$

identical to the combined $M/M/1$ result of Model-A (1.3).

4.1.3 Class-2 mean queue length by subtraction

Subtracting (4.2) from (4.3),

$$\mathbb{E}[N_2] = \frac{\rho^2}{1 - \rho} - \frac{\rho \rho_B}{1 - \rho_B} = \frac{\rho(\rho - \rho_B)}{(1 - \rho)(1 - \rho_B)}. \quad (4.4)$$

The excess load admits a transparent decomposition,

$$\rho - \rho_B = \frac{\lambda_1 + \lambda_2}{\mu} - \frac{\lambda_1}{\mu + \gamma_1} = \frac{(\lambda_1 + \lambda_2)(\mu + \gamma_1) - \lambda_1 \mu}{\mu(\mu + \gamma_1)} = \frac{\lambda_2 \mu + \lambda_1 \gamma_1 + \lambda_2 \gamma_1}{\mu(\mu + \gamma_1)} = \rho_2 + \frac{\lambda_1 \gamma_1}{\mu(\mu + \gamma_1)},$$

so $\rho - \rho_B$ exceeds the bare class-2 load ρ_2 precisely by the load that jockeying transfers from Queue 1 into Queue 2. At $\gamma_1 = 0$ it collapses to ρ_2 and (4.4) recovers Model-A's $\mathbb{E}[N_2] = \rho \rho_2 / [(1 - \rho_1)(1 - \rho)]$.

4.1.4 Mean waiting times via Little's Law

We apply **Little's Law** to each queue separately, recalling that N_i counts only the customers *waiting*, so $\mathbb{E}[N_i] = \Lambda_i \mathbb{E}[W_i]$ with Λ_i the long-run rate at which customers *enter* Queue i and $\mathbb{E}[W_i]$ their mean residence time in that queue (from entry until they leave it, by service entry or by jockeying).

Every class-1 customer enters Queue 1 exactly once, by exogenous arrival, so $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{\rho \rho_B}{\lambda_1(1 - \rho_B)} = \frac{\rho}{\mu + \gamma_1 - \lambda_1}.$$

Compared with Model-A's $\mathbb{E}[W_1] = \rho/(\mu - \lambda_1)$ (1.4), the denominator is enlarged by γ_1 : head-of-line jockeying drains the priority queue faster and shortens class-1 waits.

Queue 2 receives a *mixed* input: the exogenous Poisson stream at rate λ_2 together with the jockeying stream. The latter fires at rate γ_1 whenever $N_1 \geq 1$, so its long-run rate is $\gamma_1 \mathbb{P}(N_1 \geq 1)$. The probability that a class-1 customer is waiting follows from the geometric marginal (4.1) by subtracting the $N_1 = 0$ mass from the busy mass,

$$\mathbb{P}(N_1 \geq 1) = P(1, 1) - P(0, 1) = \rho - \rho(1 - \rho_B) = \rho \rho_B,$$

since $P(1, 1) = 1 - \pi_0 = \rho$ and $P(0, 1) = P_y(1) = \rho(1 - \rho_B)$. The total entrance rate into Queue 2 is therefore $\Lambda_2 = \lambda_2 + \gamma_1 \rho \rho_B$, and Little's Law gives the mean residence time per Queue-2 entrant,

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\Lambda_2} = \frac{\rho(\rho - \rho_B)}{(1 - \rho)(1 - \rho_B)(\lambda_2 + \gamma_1 \rho \rho_B)}. \quad (4.5)$$

As $\gamma_1 \rightarrow 0^+$ the jockeying stream disappears, $\Lambda_2 \rightarrow \lambda_2$ and $\rho_B \rightarrow \rho_1$, and (4.5) recovers Model-A's $\mathbb{E}[W_2] = \rho/[\mu(1 - \rho_1)(1 - \rho)]$ (1.5).

5 Model- C_2^H (head-of-line abandonment)

5.1 Mean queue lengths and mean waiting times

5.1.1 Class-1 marginal PGF and mean queue length

The class-1 marginal is the geometric law (??); pinning the constant by $P(1, 1) = 1 - \pi_0$ as in (??) gives $P(x, 1) = (1 - \pi_0)(1 - \rho_C)/(1 - \rho_C x)$. Differentiating and evaluating at $x = 1$,

$$\mathbb{E}[N_1] = \frac{d}{dx} P(x, 1) \Big|_{x=1} = \frac{(1 - \pi_0) \rho_C}{1 - \rho_C}, \quad \rho_C = \frac{\lambda_1}{\mu + \theta_1}, \quad (5.1)$$

with $1 - \pi_0 = \lambda(\mu + \theta_1)/[\mu(\mu + \theta_1) + \lambda_1\theta_1]$ from Corollary ?? . As in Model- B_2^H , the priority queue behaves as an $M/M/1$ queue with effective clearance rate $\mu + \theta_1$, the only difference from Model- B_2^H 's (4.2) being the prefactor $1 - \pi_0$, which here is no longer equal to ρ because abandonment alters the idle probability.

5.1.2 Class-2 mean queue length

Abandonment is a true departure and so breaks the *conservation of the total customer count* that underlies the diagonal argument: in Model- B_2^H the jockeying factor $\gamma_1 y(x - y)$ vanishes on the diagonal $x = y$, whereas here the abandonment factor $\theta_1 y(x - 1)$ does not. Concretely, setting $x = y = z$ in (??) leaves the term $\theta_1 z(z - 1)P_y(z)$ on the right-hand side, so the diagonal equation no longer closes and the route that furnished $\mathbb{E}[N_1] + \mathbb{E}[N_2]$ for Model- B_2^H is unavailable. We instead differentiate the class-2 marginal. From the marginal relation (??), $P(1, y) = \mu[P_y(y) - \pi(0, 0)]/(\lambda_2 y)$, and since $\mathbb{E}[N_2] = \frac{d}{dy}P(1, y)|_{y=1}$,

$$\mathbb{E}[N_2] = \frac{\mu}{\lambda_2} P'_y(1) - (1 - \pi_0), \quad (5.2)$$

where we used $\frac{\mu}{\lambda_2}[P_y(1) - \pi(0, 0)] = P(1, 1) = 1 - \pi_0$. It remains to compute $P'_y(1)$ from (??). Both the numerator $\mu x^*(y)(1 - y)\pi(0, 0)$ and the denominator $\Delta(y) = (\mu + \theta_1 y)x^*(y) - y(\mu + \theta_1)$ vanish at $y = 1$ (since $x^*(1) = 1$ gives $\Delta(1) = (\mu + \theta_1) - (\mu + \theta_1) = 0$), so $P_y(y)$ is a 0/0 form there. Writing $N(y)$ and $\Delta(y)$ for numerator and denominator, a second-order Taylor expansion about $y = 1$ gives the value and the slope

$$P_y(1) = \frac{N'(1)}{\Delta'(1)} = (1 - \pi_0)(1 - \rho_C), \quad P'_y(1) = \frac{N''(1)\Delta'(1) - N'(1)\Delta''(1)}{2\Delta'(1)^2},$$

which is why both the first and the second root sensitivities of x^* at $y = 1$ are needed, namely (??) together with

$$\frac{d^2}{dy^2}x^*(1) = \frac{2(\mu + \theta_1)\lambda_2^2}{(\mu + \theta_1 - \lambda_1)^3}$$

(the latter obtained by differentiating $x^*(y)$ in (??) twice). Substituting these into $P'_y(1)$, then into (5.2), and simplifying yields the closed form

$$\mathbb{E}[N_2] = \frac{\lambda_2 \lambda (\mu + \theta_1) [(\mu + \theta_1)^2 - \lambda_1 \theta_1]}{(\mu + \theta_1 - \lambda_1) [\mu(\mu + \theta_1) + \lambda_1 \theta_1] [(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1]}, \quad (5.3)$$

with $\lambda = \lambda_1 + \lambda_2$. The last bracket in the denominator is exactly the stability numerator of π_0 from (??), so $\mathbb{E}[N_2]$ is finite precisely under the stability condition (??) and diverges as $\pi_0 \rightarrow 0^+$. As $\theta_1 \rightarrow 0^+$ the right-hand side reduces to $\lambda_2 \lambda / [(\mu - \lambda_1)(\mu - \lambda)] = \rho \rho_2 / [(1 - \rho_1)(1 - \rho)]$, recovering Model- A 's (1.3).

5.1.3 Mean waiting times via Little's Law

We apply **Little's Law** to each queue. Every class-1 customer enters Queue 1 once by exogenous arrival and later leaves it either by entering service or by abandoning from the head; in both cases it leaves Queue 1, so the entrance rate is $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{(1 - \pi_0)\rho_C}{\lambda_1(1 - \rho_C)}.$$

Here $\mathbb{E}[W_1]$ is the mean time a class-1 customer spends *waiting* in Queue 1, averaged over all class-1 arrivals, whether they are eventually served or abandon. Queue 2, by contrast, receives no jockeying input and its customers never abandon ($\theta_2 = 0$), so every class-2 arrival is eventually served and the entrance rate is simply $\Lambda_2 = \lambda_2$:

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2}$$

with $\mathbb{E}[N_2]$ from (5.3), the λ_2 cancelling against its numerator. Both means reduce to the corresponding Model- A expressions as $\theta_1 \rightarrow 0^+$.

6 Loss fractions: the cost of abandonment

6.0.1 The cost of abandonment in lost customers

The headline gains of C_2 — shorter queues, larger idle fraction, lower priority waiting time — are bought with lost customers, and the price is exact in closed form. The server delivers carried work at rate $\mu(1 - \pi_0)$ and the offered rate is $\lambda_1 + \lambda_2 = \mu\rho$. Since in steady state every arrival is either served or abandons, flow conservation forces the total abandonment rate to equal the *idle excess* that abandonment creates over the no-attribution baseline,

$$\Lambda_{\text{loss}} = \mu\rho - \mu(1 - \pi_0) = \mu(\pi_0 - (1 - \rho)), \quad (6.1)$$

the strictly positive gap $\pi_0 - (1 - \rho) > 0$ of Corollary ???. The identity (6.1) holds for either abandonment rate — the full-rate mechanism realises $\Lambda_{\text{loss}} = \theta_1 \mathbb{E}[N_1]$ and the head-of-line mechanism $\Lambda_{\text{loss}} = \theta_1 \mathbb{P}(N_1 \geq 1)$ — because it uses only the carried rate $\mu(1 - \pi_0)$. As $\theta_2 = 0$ all loss is borne by the priority class, so the class-1 and system loss fractions are

$$L_1 = \frac{\Lambda_{\text{loss}}}{\lambda_1} = \frac{\pi_0 - (1 - \rho)}{\rho_1}, \quad L = \frac{\Lambda_{\text{loss}}}{\lambda_1 + \lambda_2} = \frac{\pi_0 - (1 - \rho)}{\rho}, \quad (6.2)$$

both closed-form through the C_2 idle probability π_0 , and both $\rightarrow 0$ as $\theta_1 \rightarrow 0^+$ (where $\pi_0 \rightarrow 1 - \rho$), recovering the lossless Model-A limit. At $\rho = 0.70$, $\theta_1 = 0.5$ this gives $L_1 = 0.232$ — 23.2% of the priority class is shed — against a system loss of only $L = 0.099$, the difference reflecting that the non-priority class never abandons. Table 1 collects the loss fractions at the three reference loads for the full-rate C_2 and the head-of-line C_2^H ; the head-of-line valve, capped at θ_1 rather than scaling as $\theta_1 n_1$, sheds uniformly less at every load.

Table 1: Loss fractions under class-1 abandonment ($\theta_1 = 0.5$, $\lambda_1:\lambda_2 = 3:4$, $\mu = 1$), from (6.2) with the model’s idle probability π_0 . L_1 is the fraction of class-1 arrivals shed; L is the system-wide fraction. The head-of-line model C_2^H sheds uniformly less than the full-rate C_2 .

ρ	Model- C_2		Model- C_2^H	
	L_1	L	L_1	L
0.50	0.166	0.071	0.156	0.067
0.70	0.232	0.099	0.212	0.091
0.90	0.297	0.127	0.266	0.114

The price in lost customers. The loss fraction is the strongest applied statement this thesis makes, and it is blunt: at $\rho = 0.70$ with $\theta_1 = 0.5$ a fraction $L_1 = 0.232$ of class-1 arrivals — almost one in four — *never reach the server at all*. The priority waiting time collapses from $\mathbb{E}[W_1] = 1.0000$ in Model-A to 0.4639 under C_2 — but partly because roughly a quarter of the queue walks away, not because it is served faster: the conditional mean time-to-service of *served* class-1 customers is 0.3863, against the count-based 0.4639. In the healthcare setting an “abandonment” is a patient who deteriorates past the point of benefit or leaves the list by death or transfer, so L_1 estimates the high-priority demand lost to delay. A waiting-time figure that reads as an improvement can therefore encode a worsening of the outcome that matters most, so any operational use of a C_2 -type policy must report L_1 *alongside* $\mathbb{E}[W_1]$, never in its place.

The loss fraction rises with load — abandonment is most active precisely where the priority queue is longest — but never approaches one: even at $\rho = 0.90$ roughly seventy per cent of class-1 customers are still served. Figure 1 maps where abandonment is most effective as a congestion control: the ratio $\mathbb{E}[N_1]_{C_2}/\mathbb{E}[N_1]_A$ is everywhere below unity and smallest in the high- ρ , high- θ_1 corner, where the Model-A baseline is largest and the valve acts hardest.

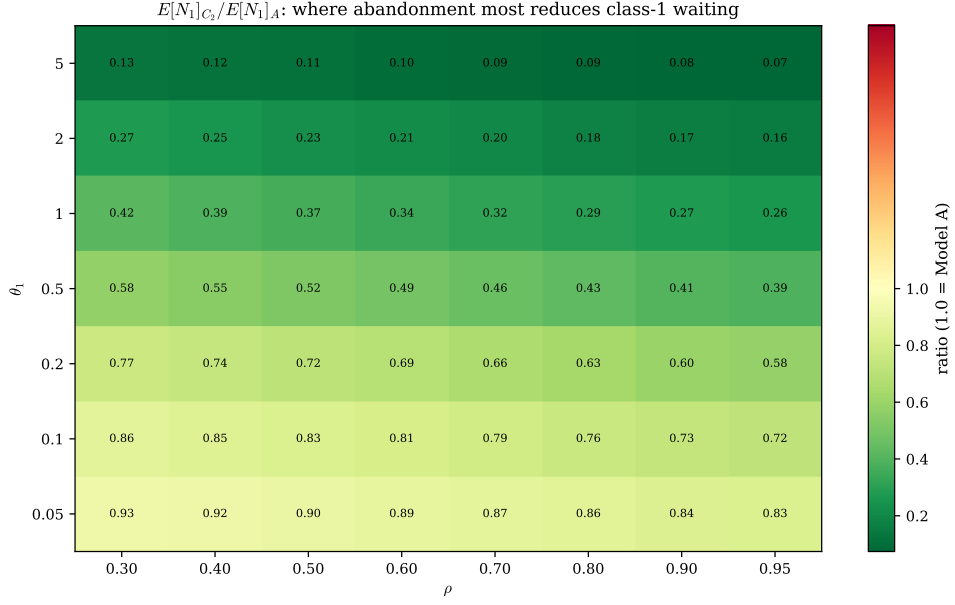


Figure 1: Ratio $\mathbb{E}[N_1]_{C_2}/\mathbb{E}[N_1]_A$ over the (ρ, θ_1) plane; the diverging scale is centred at 1.0 (the Model-A value). *Colour convention* (shared with the validity map, Figure ??): **green** marks the favourable regime—ratio < 1 , where abandonment shortens the priority queue—and **red** the unfavourable regime, ratio > 1 ; yellow is the break-even Model-A value. Abandonment delivers its largest relative reduction in the priority queue at high load and high θ_1 (bottom-right region), exactly where the baseline backlog is most severe.

Read across the panel, the ratio varies far more strongly down the θ_1 axis than along the ρ axis, so it is the *valve strength*, not the offered load, that is the first-order control on how much abandonment helps the priority class. Further θ_1 -sweeps are deferred to Appendix ??.